

Does a Music Transformer Know What Key It Is In?

Causal Interventions on an Emergent Tonal Representation

Anonymous ICASSP 2027 submission

Abstract—Musical attributes can be decoded from the activations of music generation models, but decodability does not imply that generation *uses* the decoded representation. We test use directly: in Transformers trained on symbolic music whose vocabulary holds absolute pitch and time only, we replace the low-dimensional key component of the hidden state with another key’s while the musical prompt stays identical. On a fully held-out confirmatory set, with the intervention layer, subspace, metrics and tests all frozen in advance, the edit moves generation to the injected key where rank- and norm-matched random edits do not ([TBD (run required): held-out sweep running]), within a musicality budget frozen before any edit ran. The effect precedes feedback from generated notes: before a single new note is sampled, the edit shifts the next-pitch distribution toward the injected key (+0.26 probability mass; matched random +0.02; 12/12 keys, Holm-corrected). The representation is broadly decodable across layers but causally effective only in a narrow middle band, and edits restricted to pitch-token positions are causally null—so the effect is not a trivial write to pitch decisions, and where a state is readable is a weak guide to where it is used. Probe and intervention transfer to a public checkpoint trained on real music (7.8× the matched control, 12/12 keys). We conclude that a music Transformer computes a manipulable key representation that its generation process causally consults.

Index Terms—music generation, interpretability, causal intervention, tonality, probing

I. INTRODUCTION

Systems that compose music write convincingly tonal pieces [1], [2], [3]: they settle into a home key, build tension away from it, and return. Probing studies show that musical attributes, key among them, can be *read out* of such models’ activations by trained classifiers [4]. But a signal can be decodable from a system’s state without the system consulting it when it acts [5], [6]. Whether a music model’s generation process *uses* an internal key representation has not been tested.

The test exists in other domains. A model trained only to predict Othello moves tracks the board those moves imply, and editing that internal board changes which moves it treats as legal [7], [8]; the same causal standard has reached chess, mazes, and RL environments [9], [10], [11]. Music

has probes, and it has steering—inference-time interventions adapted from language models [12], [13] that push attributes along learned directions [14], [15]—but steering shows that a direction moves an attribute, not that the model maintains a representation whose *value* can be set: removed, replaced by another value, and then acted on.

We run that test with musical key as the variable. Key is defined outside the model by theory and psychological measurement [16], must be computed by integrating pitch evidence over time (the vocabulary names no key), and its scale gives an objective success measure with no learned judge. We replace the key component of one layer’s activations with a donor key’s and ask what the model composes next—under a design in which every choice (layer, subspace, metrics, tests) was frozen before the confirmatory data were seen, an intervention-free logit-level test rules out feedback from generated notes, and token-type ablations rule out a trivial write to pitch decisions. We release the testbed—leak-free tokenizer, corpus with exact per-token key labels, frozen protocol—as a benchmark.

II. METHOD

A. Data, models, protocol

We need the true key at every note, so we generate training data. **D-SYN** comes from a functional-harmony state machine (four-part voicing plus optional melody; 200k/10k/10k pieces of 278–508 tokens; 0–3 modulations each). Every token carries a true key label $\kappa(t)$ (24 classes), **which never reaches the model**: the vocabulary (124 types) holds only BAR, POS, absolute-MIDI PITCH, DUR and structural symbols, enforced by a unit test. **M-CTRL** is a GPT-2 style decoder [17], [18] (8 layers, $d=512$, ~ 25 M parameters), trained with and without transposition augmentation $\times 3$ seeds. **M-REF** shares the architecture but a different seed and data slice; it only judges whether edited music still sounds like music, so no model grades its own work. All models clear a preset quality gate (validation top-1 0.878–0.881 vs. the 0.434 of a most-frequent-token predictor). Decision rules, thresholds and controls were frozen before experiments ran; deviations are logged and negative results reported.

B. Probing with controls

Linear probes [19], [6] map the residual stream $h_\ell(t)$ to $\kappa(t)$, split by piece, read at uniformly sampled positions. Three control families bound them: **C1** label permutations (their score on true keys is the *selectivity floor* [5], subtracted

from every margin); **C2** an untrained model; **C3** input-side baselines that name the key from the notes alone—logistic regression on pitch-class histograms and Krumhansl-Schmuckler (KS) matching [20], [21] over windows $W \in \{8, \dots, 64\}$, extended in Sec. III-A to the full prefix. The frozen rule **DR-H1**: the probe’s selectivity-corrected F_1 must beat the best C3 with a piece-level BCa [22] 95% CI excluding zero.

C. Intervention

From bar 9 of an 8-bar key-stable prompt, a sustained edit replaces the key component at one layer at every generated position, following activation patching [23], [24]: $h \leftarrow h - P_V h + P_V \mu_{\kappa^*}$, with V an orthonormal key subspace, $P_V = VV^\top$, and μ_{κ^*} the class-mean activation of the donor key. **The prompt never changes.** V is built three ways (probe row space **V-PROBE**; class-mean span **V-MEAN**; optimized **V-DAS** [25]); four controls bound the result: **K1**, a rank- and norm-matched random subspace (measured $\|\Delta\|$ within 1% of the edit’s); **K2**, a sham edit that must reproduce clean output bit for bit (it does, 100/100, re-verified on every prompt set); **K3**, another layer’s subspace; **K4**, transposing the prompt itself—the behavioral ceiling. Continuations are scored by the *target-key rate* (TKR: KS estimate equals the injected key, reported strictly) and the *in-key rate* (IKR: fraction of pitches in a given scale, no estimator). A musicality budget $\delta_{\text{PPL}} = 0.613$ —the 90th percentile of M-REF’s perplexity rise over 7,989 *natural* modulations, frozen before any edit—discards edits that damage the music more than a real modulation; all headline rates are *guarded*.

D. Confirmatory design

Our first sweep compared 8 layers $\times$ 3 subspaces on 100 prompts and evaluated the best cell (V-PROBE, L4) on those same prompts—and those prompts lie inside the probe’s training pool. Holm correction covered the 12 targets, not the selection. We therefore treat that sweep as *selection*, and add a *confirmatory* phase: 100 fresh prompts from a region of the test split untouched by probe training and by every prior run; layer, subspace, timing, targets, controls, guard, metrics, tests, seeds and sample size all frozen in a version-controlled document committed before the first confirmatory number existed. No layer or subspace search is repeated. Identity cells (injected key equals the prompt’s key) are excluded from the primary analysis and reported as a sanity check. All intervention choices and evaluation criteria were frozen before evaluating the held-out set.

III. RESULTS

A. The key is decodable beyond strong input baselines

At L4 the probe reaches macro- F_1 0.927 against the best pre-registered input baseline 0.790 and a selectivity floor of 0.032: corrected margin +0.105 (BCa 95% CI [0.091, 0.114]), positive in layers 1–7 of all six trained models (DR-H1 6/6), negative at the raw embedding (−0.20), 0.243 for an untrained model. Because the probe attends the full prefix while C3 saw $W \leq 64$ tokens, we extended the baseline to the full prefix plus recency-decayed and short+long-concatenated variants: the window curve turns over at $W=96$ (0.801) and falls to 0.752 at $W=512$ —under modulations, stale evidence poisons a histogram—and against the strongest variant (0.824) the margin remains positive, +0.071 [0.056, 0.081], with all twelve size/seed verdicts unchanged. We say “more linearly accessible than these pitch-statistical baselines,” not

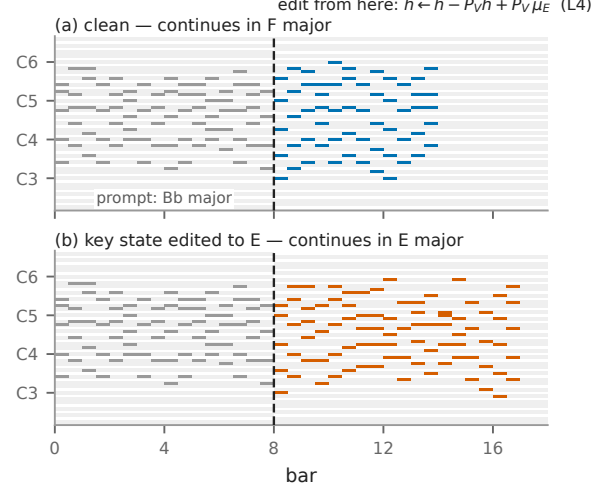


Fig. 1. The claim in one example (real model output, fixed sampling seed). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 on. **The input does not change**; the continuation follows the injected internal value. Shading marks the scale of the key in force.

“beyond all conceivable surface functions”: hidden states are themselves functions of the input.

B. Main causal result: held-out, pre-specified

Selection phase (the original sweep; its prompts overlap probe training): V-PROBE at L4 was the best in-budget cell, guarded strict TKR 0.378 vs. its layer-matched K1 0.075 (5.0 $\times$; 12/12 targets, Holm-corrected $p < 10^{-4}$; guard pass 87%; V-DAS 0.213; V-MEAN destroys music at deep layers, in-budget $\leq 5\%$; 58% of the K4 ceiling). Raw pitch content agrees without the estimator: $\text{IKR}_{\kappa^*} = 0.937$ vs. $\text{IKR}_{\text{src}} = 0.646$.

Confirmatory phase (held-out prompts; everything frozen at commit time; identity cells excluded): **[TBD (run required): held-out confirmatory sweep executing — per-target guarded TKR, K1, rank-biserial r , Holm-corrected p , BCa CI, guard rate, and identity sanity numbers land here]**.

Specificity. The edit does not merely leave the source key; it lands on the injected one. **[TBD (run required): confirmatory specificity: diagonal mass, fifth-neighbor mass, source-retention, edit vs. K1]**. In the selection-phase data, success falls gently with circle-of-fifths distance (0.395 at one step, 0.280 at the tritone), as moving a key should.

C. The effect precedes any generated note

A sustained edit leaves a loophole: perhaps the edit perturbs a few early notes and the model then re-estimates the key from its own output. We close it at the logit level. Each held-out prompt is extended by a teacher-forced, key-neutral structural prefix (**BAR, POS_1**); the edit is applied from the bar-9 boundary; *no token is ever sampled*. We read the next-token distribution over pitch tokens and score the frozen metrics P_{target} (probability mass diatonic to the injected key) and $\Delta_{\text{key}} = \log P_{\text{target}} - \log P_{\text{source}}$. The edit shifts Δ_{key} by +0.695 over clean; matched random edits shift it +0.041.

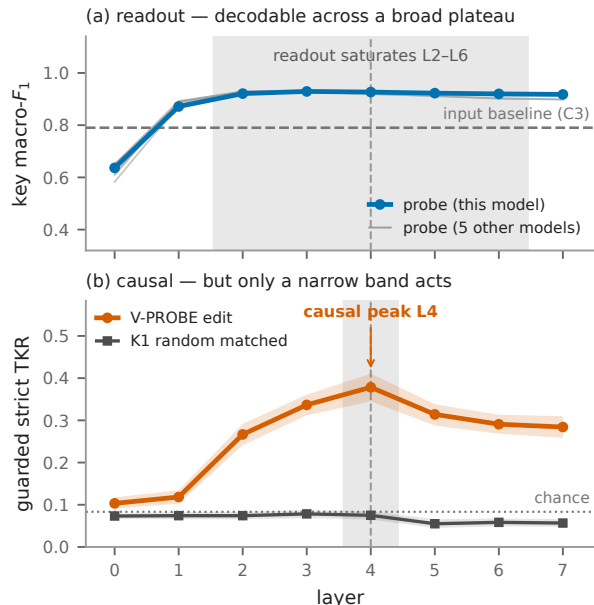


Fig. 2. (a) Probe macro- F_1 per layer, all six models (thin lines; seed 0 highlighted): broadly readable. (b) Causal effect of the edit vs. the K1 matched control (bands: prompt-level bootstrap 95% CI): narrowly usable. Selection-phase data; the confirmatory run fixes L4 and does not re-search.

Target-scale mass rises $+0.255$ from a clean baseline of 0.547 (K1: $+0.017$). The separation holds in 12/12 keys (Holm-corrected worst $p=5 \times 10^{-16}$; rank-biserial $r \geq 0.99$). **The intervention changes the model’s next musical decision before any generated-note feedback can occur.**

D. Where it is readable is not where it is used

Two dissociations locate the effect. *Across layers*: the probe saturates broadly (macro- F_1 0.920–0.929 across L2–L6), but editing each layer in turn and subtracting the matched control gives a single mid-network peak (0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7; best-vs-median CI [0.038, 0.109] excludes zero)—and near-zero efficacy at L0–L1 where the probe already reads the key well (Fig. 2). The peak’s index is not sacred—the second seed replicates every verdict with its peak at L2—so we claim middle-layer concentration, not a landmark layer. *Across token types*: restricting the write to PITCH positions is causally null (selection-phase guarded TKR 0.077, at the K1 floor, music untouched), while BAR/DUR positions alone carry 69% of the full effect and no restriction improves the guard rate. **[TBD (run required): confirmatory token-type arms (all / pitch / bar-dur) land here]**. An edit that acted by trivially steering pitch logits would show the opposite pattern. Where a representation is linearly decodable is a weak guide to where it is causally used.

E. A public model trained on real music

We probe and edit the Anticipatory Music Transformer [26] (small; trained on Lakh MIDI [27], MetaMIDI [28], FMA [29]) on 300 Bach chorales [1], [30] with human local-key labels [31]. Probes here read at the positions where the model is about to choose a pitch—the convention the (time, duration, note) encoding forces, unlike

the uniform sampling of Sec. III-A; the supplementary shows the verdict inverts if one reads after the note instead. Its probe beats the surface baselines (corrected margin $+0.196$, CI [0.132, 0.260]) where our same-shaped synthetic-trained model does not ($+0.034$, CI crosses zero); the C3 baselines land within 0.003 of each other, confirming matched setups. We note the comparison varies training data and tokenizer and training procedure; it isolates “real-music training pipeline,” not data alone. The intervention transfers: with the budget re-frozen from natural chorale modulations and a *different* public checkpoint judging, the edit reaches $7.8\times$ the matched control (guarded 0.479 vs. 0.061; 12/12 targets; guard 99.7%; IKR 0.890 vs. 0.753). Its causal peak (L8) again differs from its probe peak (L10). One first-run finding did not survive its control and we retract it: an apparent “key prior” ($F\sharp$ unreachable; success tracking corpus key frequency, Spearman $+0.91$) was manufactured by estimating edit targets from the same imbalanced corpus— $F\sharp$ never occurs, so its target was a zero vector. Re-estimating from a key-balanced corpus lifts $F\sharp$ to 0.617 and collapses the correlation to a non-significant $+0.41$: a portable caution for activation editing.

F. What the intervention does not show

Three bounded claims, each measured. *Capacity*: across 0.5M–86M parameters next-token accuracy is flat, yet the smallest model has no decodable key state and its edits wreck the music (90% out of budget); manipulability peaks at 26M (full sweep in the supplementary). *Persistence*: a one-shot installation shifts pitch content durably, but splicing the installed bar’s *tokens* reproduces the entire effect—the representation is re-estimated from evidence at every step, causally consulted but not a persistent latent memory independent of emitted notes. *Geometry*: the twelve tonic directions arrange on a circle of fifths and transposition acts as an approximately linear map, but not a cyclic group (pre-registered H2b not supported; details in the supplementary).

IV. DISCUSSION AND LIMITATIONS

Within this training condition, the evidence supports a precise claim: the model computes an internal representation of musical key that is causally consulted during generation—manipulable, target-specific, effective before feedback, concentrated in middle layers and at structural token positions. “World model” is one interpretation of that bundle; we use it only here, and only as interpretation. The gaps are informative: the edit reaches 58% of the behavioral ceiling, fails where the state is most readable (L0–L1, PITCH positions), and K3 shows the middle-stack subspace is shared rather than re-learned per depth. *Bounds*. Major targets only (minor is pre-registered but unrun). The KS estimator confuses fifth-related keys; IKR triangulates it. The likelihood-based guard measures distributional damage, not musicality itself; no listening test was run. The public-model causal result is one checkpoint at one size. The confirmatory phase covers our own model; the public-model intervention retains its original design (layer chosen on disjoint prompts, budget frozen before evaluation, but no second held-out round). Supplementary material carries the capacity sweep, geometry, persistence, specificity matrices, per-seed variants, and the balanced re-estimation study.

V. CONCLUSION

A music Transformer does not merely expose correlates of key to a probe. Under a pre-registered, selection-corrected

design, replacing a rank-24 component of its hidden state moves composition to the injected key—before any generated note can feed back, specifically toward the target, within a frozen musicality budget, and reproducibly across seeds and on a public model trained on real music. Decodability and causal use dissociate across layers and token types; reading a representation is not evidence that the model consults it. The testbed, labels, and frozen protocols are released as a benchmark.

Reproducibility. Anonymized code, data generator, pre-registration and freeze documents, audio demos: <https://anonymous.4open.science/r/TBD>. [TBD (run required): create before submission.]

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